

suffered setbacks. In 1997, the University's stake was worth only \$4 million. Seragen was eventually bought by Ligand Pharmaceuticals Inc. in 1998 for \$30 million.

Norway's sovereign wealth fund, in contrast, was created precisely to reap the gains from diversification. Through its sovereign wealth fund, Norway swaps a highly concentrated asset—oil—into a diversified financial portfolio and thus improves its risk–return trade-off.

2.6. HOME BIAS

Mean-variance investing prescribes that investors should never hold a 100% U.S. portfolio. Many investors do not take advantage of the benefits of international diversification and instead hold only domestic assets. This is the *home bias* puzzle.¹³

One measure of home bias is the extent to which domestic investors' equity holdings differ from the world equity market portfolio. The world market is a diversified portfolio and, according to the CAPM, which adds equilibrium conditions to the mean-investing framework (see chapter 6), it is the optimal portfolio which investors should hold. (I show below, in section 5, that the market portfolio performs well in a horse race with other mean-variance diversified portfolios.) Ahearne, Grier, and Warnock (2004) contrast the 12% proportion of foreign equities in U.S. investors' portfolios with the approximate 50% share of foreign equities in the world market portfolio. Home bias is not just a U.S. phenomenon. Fidora, Fratzscher, and Thimann (2007) report the share of foreign assets in U.K. investors' portfolios is 30% compared to the non-U.K. weight of 92% in the world market portfolio. Japan is even more home biased, with domestic investors owning fewer than 10% foreign equities while the non-Japanese positions in the world market portfolio are greater than 90%.

What accounts for home bias?

1. Correlations vary over time

The diversification benefits of international investing depend on there being low correlations (or at least correlations that are not near one in absolute value). If correlations of international equity returns increase when there are global crashes—such as in 1987 (stock market crash), 1998 (emerging market crisis brought about by Russia's default), and the financial crisis over 2008–2009—then international diversification benefits may not be forthcoming when investors most desire them. Correlations of international stock markets do increase, and diversification benefits correspondingly decrease, during global bear markets. But Ang and Bekaert (2002) show that there

¹³ See Karolyi and Stulz (2003) and Lewis (2011) for literature summaries.